Data Science (Optativa I)

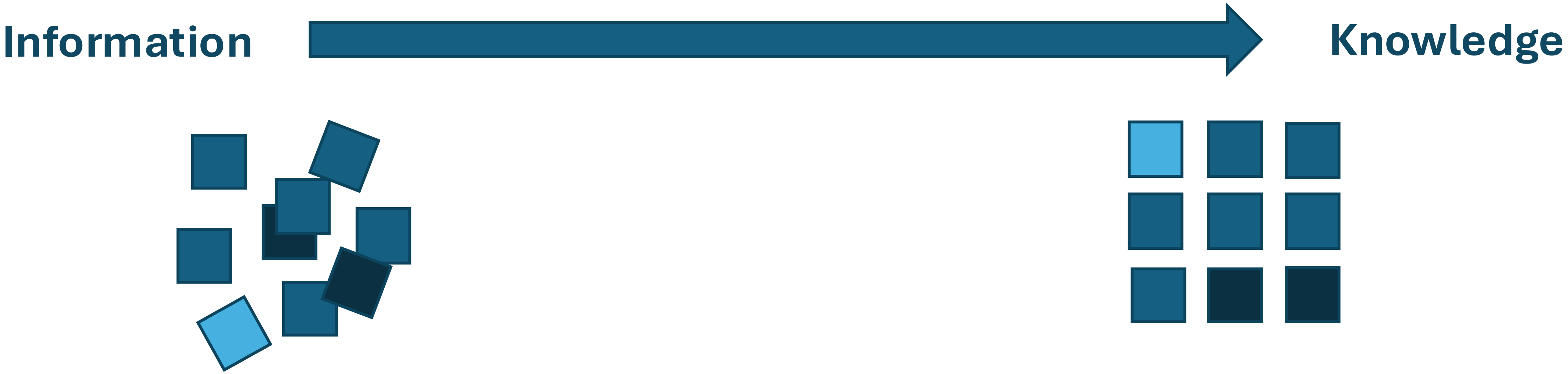
Unit 2 – Practical Assignment: Data Cleaning and Preparation

Ciência da Computação
Prof. Maiko Spiess

1. Previously...

2. Practical assigment

Previously...



- Organize
- Classify
- Describe
- Identify relationships
- Test hypotheses

Specific — "Low-level" perspective

General — "High-level" perspective

Types of **variables** in programming languages

Types of **data** in statistics and scientific research

Types of data used by **organizations**

Types of Data in Data Science

Data Science operates across these levels, combining computational representations of data with statistical classifications and real-world analytical applications.

How Can We Record the Same Type of Information? (e.g. social class)

Different representations of the same phenomenon produce different types of data, which in turn influence the analytical methods that can be applied!

Theoretical categories ⟷ **Exact measurement**

	Social stratification levels	**Income brackets**	
Bourgeoisie **Proletariat**	**Upper class** **Middle class** **Lower class**	**A: R$ 20,900 or more** **B: R$ 10,450 – R$ 20,900** **C: R$ 4,180 – R$ 10,450** **D: R$ 2,090 – R$ 4,180** **E: up to R$ 2,090**	**Total monthly income (in Brazilian reais)**

In the context of Data Science, the expression **"garbage in, garbage out" (GIGO)** emphasizes that the quality of analytical results is directly proportional to the quality of the input data.

In other words, low-quality data—such as inaccurate, incomplete, or inconsistent information—inevitably leads to flawed or misleading results.

This principle highlights the importance of ensuring the integrity and accuracy of the data used in analytical processes in order to obtain **valid** and **reliable** conclusions.

Data preparation

Data Preparation and Data Wrangling: Conceptual Distinction

Core idea: Data preparation is a broader methodological stage; data wrangling is its operational core.

Definitions:

- **Data Preparation:** The end-to-end process of making data suitable for analysis or modeling. It includes selection, transformation, encoding, validation, and partitioning.
- **Data Wrangling (Data Munging):** The technical process of cleaning, transforming, and restructuring raw data into a usable format.

Missing and Duplicate Values

The identification and treatment of missing and duplicate values are fundamental steps in data cleaning.

These issues can introduce bias, reduce statistical power, and compromise the reliability and validity of subsequent analyses.

Proper handling involves not only detecting such inconsistencies but also applying appropriate strategies—such as **imputation**, **removal**, or **deduplication**—based on the analytical context and the underlying data-generating process.

Missing Data Mechanisms

Missing data mechanisms describe the process that generates missing values in a dataset. Understanding this mechanism is essential for selecting appropriate handling strategies and avoiding biased results.

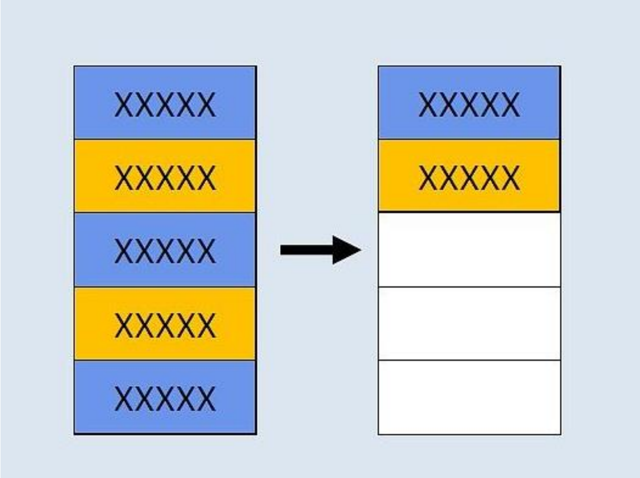
Mechanism	Definition	Bias Risk	Example
MCAR (Missing Completely at Random)	Missingness is entirely random and unrelated to any variable	Low	A survey response is missing because a random system error occurred during data recording.
MAR (Missing at Random)	Missingness depends on observed variables	Moderate	Income is missing more frequently among younger respondents, but age is observed in the dataset.
MNAR (Missing Not at Random)	Missingness depends on the missing value itself or unobserved factors	High	Individuals with very high income choose not to report it, so missingness depends on the value itself.

Removing Duplicates

Duplicate removal involves identifying and **eliminating identical or near-identical records** within a dataset. This is a critical step in data cleaning, particularly when integrating multiple data sources or performing merge operations that may inadvertently introduce duplicates.

Advantages:

- **Improved Data Quality:** Removing duplicates increases dataset reliability and accuracy, which is essential for valid downstream analysis.
- **Computational Efficiency:** Duplicate records consume storage and processing resources. Their removal can improve performance in data processing and modeling tasks.
- **Noise Reduction:** Duplicates may introduce bias or distort statistical assumptions, especially in methods that rely on independent observations.
- **Greater Consistency:** Maintaining unique records simplifies interpretation and reduces ambiguity in analytical results.



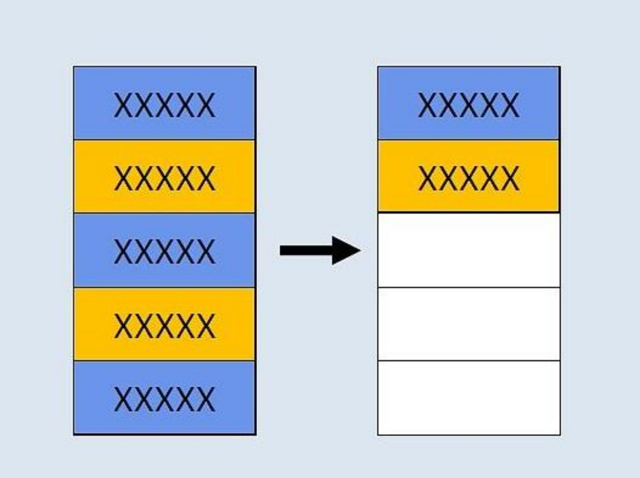
Removing Duplicates

Disadvantages

Information Loss: Apparent duplicates may reflect legitimate variations in the data. Indiscriminate removal can eliminate meaningful nuances.

Induced Bias: If duplicates are not randomly distributed, their removal may bias the dataset and affect inferential validity.

Identification Complexity: Detecting duplicates is not always trivial. Variations in names, addresses, or formats may require advanced techniques such as fuzzy matching or record linkage.



Imputation Using Mean, Median, or Mode

This approach replaces missing values with a **measure of central tendency**—such as the **mean** (arithmetic average), **median** (middle value), or **mode** (most frequent value)—computed from the observed data for the variable of interest.

Advantages:

- **Simplicity and Ease of Implementation:** Straightforward to apply and does not require complex statistical models.
- **Preservation of Sample Size:** Unlike case deletion, this method retains all observations, maintaining the dataset size and statistical power.
- **Distributional Stability (Partial):** Central tendency imputation reduces the impact of missing values on the overall distribution, although it does not fully eliminate bias.

Disadvantages:

- **Reduced Variability:** Imputation with mean, median, or mode decreases variance, potentially distorting the data distribution and affecting statistical inference.
- **Statistical Bias:** If data are not missing at random, this approach may introduce systematic bias.
- **Limitations for Data Types:**
    - Mean is only appropriate for interval/ratio data
    - Median is suitable for ordinal or skewed numerical data
    - Mode is commonly used for categorical variables, but may oversimplify variability

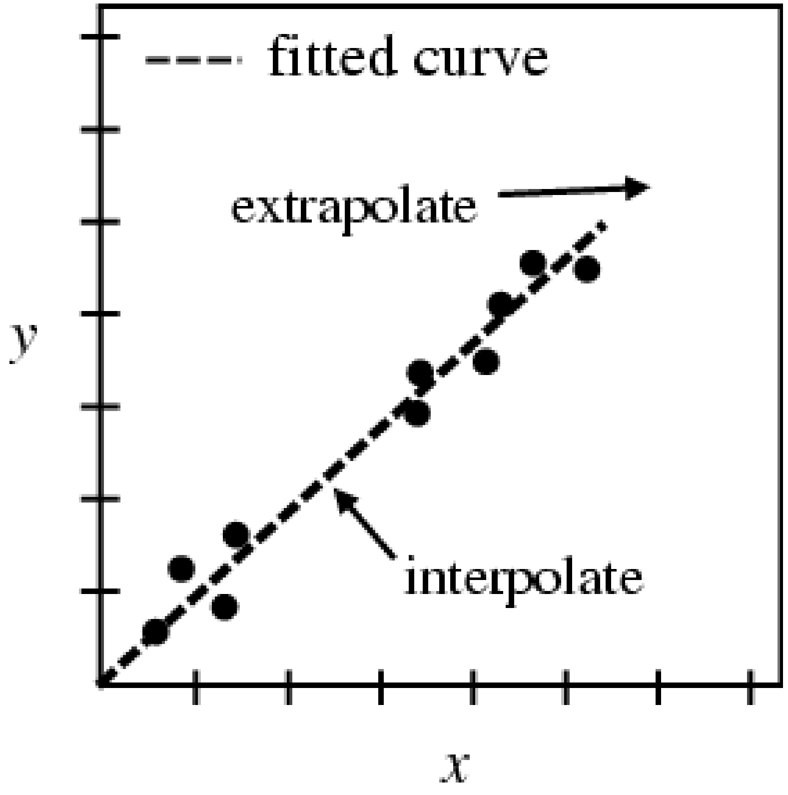
Interpolation and Extrapolation

Interpolation and extrapolation are techniques commonly used to estimate missing or unknown values in **time series or sequential data**. These methods rely on existing data points to infer values either within the observed range (interpolation) or beyond it (extrapolation).

Interpolation: Estimation of values within the range of observed data

Extrapolation: Estimation of values outside the observed data range

Interpolation is generally more reliable, as it operates within the support of observed data, whereas **extrapolation** is more uncertain and model-dependent.



Interpolation

Advantages:

- **Accuracy (Conditional):** Can provide accurate estimates when the underlying process is smooth and well-behaved.
- **Continuity Preservation:** Maintains continuity in the data, which is particularly important in time series where observations are temporally dependent.

Disadvantages:

Dependence on Observed Range: Interpolation can only estimate values within the range of available data; it cannot infer beyond observed boundaries.

Model Assumptions: Many interpolation methods (e.g., linear interpolation) assume specific functional forms (such as linearity), which may not hold in complex or non-linear data.

Extrapolation

Advantages:

- **Extensibility**: Useful when predictions are required for data points outside the observed range.
- **Long-Term Trend Analysis:** Can support forecasting and the analysis of long-term trends, particularly in time series contexts.

Disadvantages:

- **High Risk of Error:** Because it extends beyond observed data, extrapolation is more prone to error and uncertainty compared to interpolation.
- **Strong Assumptions:** Assumes that the patterns observed in the data continue beyond the observed range, which may not hold in real-world scenarios.

LOCF and LOCB

LOCF (Last Observation Carried Forward) and **LOCB (Last Observation Carried Backward)** are simple imputation techniques commonly used in time series and longitudinal data to handle missing values.

- **LOCF: fills missing values using the last observed value**
- **LOCB: fills missing values using the next observed value**

These methods are easy to implement and help preserve sample size, maintaining statistical power. They are particularly useful when the variable of interest is relatively stable over time.

Limitations:

- **Stationarity assumption:** Both methods assume that values remain stable over time, which may not hold in practice.
- **Risk of bias:** If this assumption is violated, the imputed values may distort the data.
- **Limited applicability:** Not suitable for variables with strong trends or non-linear dynamics.

	LOCF	LOCB
1	1	1
2	2	3
3	3	3
4	?	
5		
6	6	6

Machine Learning Methods for Missing Data Imputation

Machine learning techniques such as **K-Nearest Neighbors (KNN)**, **Random Forest**, and **neural networks** are widely used for imputing missing data. These approaches are more sophisticated than traditional methods (e.g., mean or median imputation), as they can capture **complex patterns** and account for relationships between **multiple variables**, leading to more accurate and context-aware imputations.

Advantages:

- **Higher Accuracy:** Able to model non-linear relationships and interactions between variables, often producing more accurate estimates.
- **Versatility:** Applicable to different data types and structures, including tabular, time series, and multivariate data.
- **Robustness to Noise:** Methods like Random Forest are relatively robust to noise and outliers.
- **Multivariate Imputation:** Can leverage multiple variables simultaneously, improving the quality of imputations.

Machine Learning Methods for Missing Data Imputation

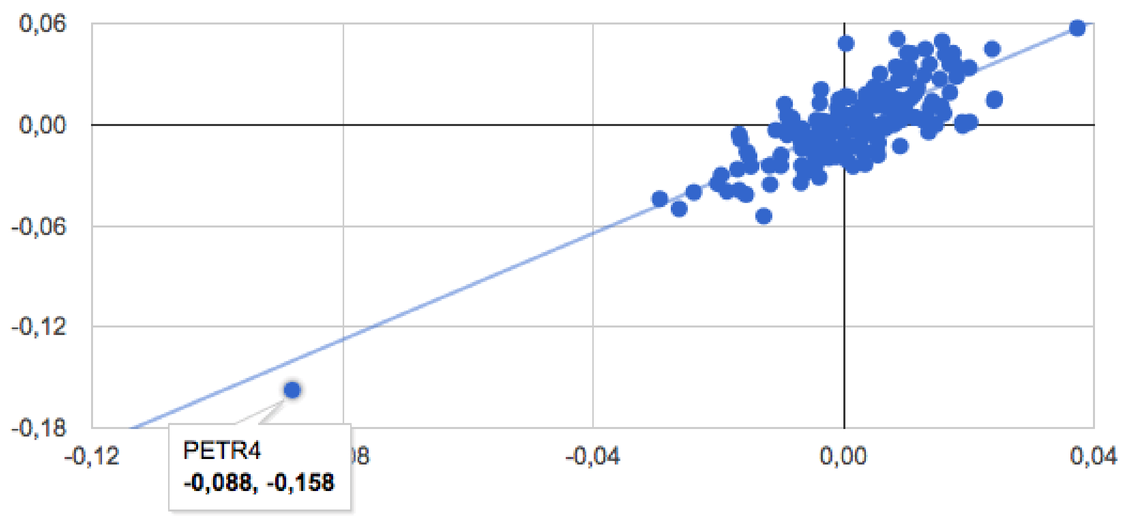
Machine learning techniques such as **K-Nearest Neighbors (KNN)**, **Random Forest**, and **neural networks** are widely used for imputing missing data. These approaches are more sophisticated than traditional methods (e.g., mean or median imputation), as they can capture **complex patterns** and account for relationships between **multiple variables**, leading to more accurate and context-aware imputations.

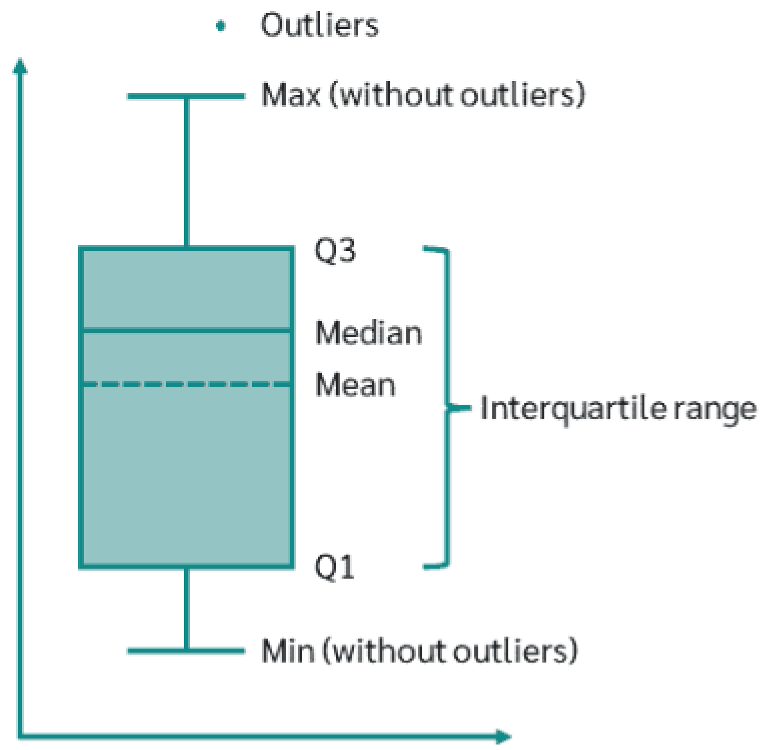
Disadvantages:

- **Computational Complexity:** Requires greater computational resources and processing time.
- **Risk of Overfitting:** Complex models may learn noise instead of true patterns, leading to unreliable imputations.
- **Limited Interpretability:** Some models (especially neural networks) act as “black boxes,” making it difficult to explain the imputation process.
- **Need for Validation:** Proper validation (e.g., cross-validation or hold-out sets) is essential to ensure reliable results.

Outliers

An outlier is an observation that deviates significantly from the overall pattern of a dataset. Outliers may arise from measurement errors, data entry issues, or genuinely rare and extreme events.





Outlier Treatment

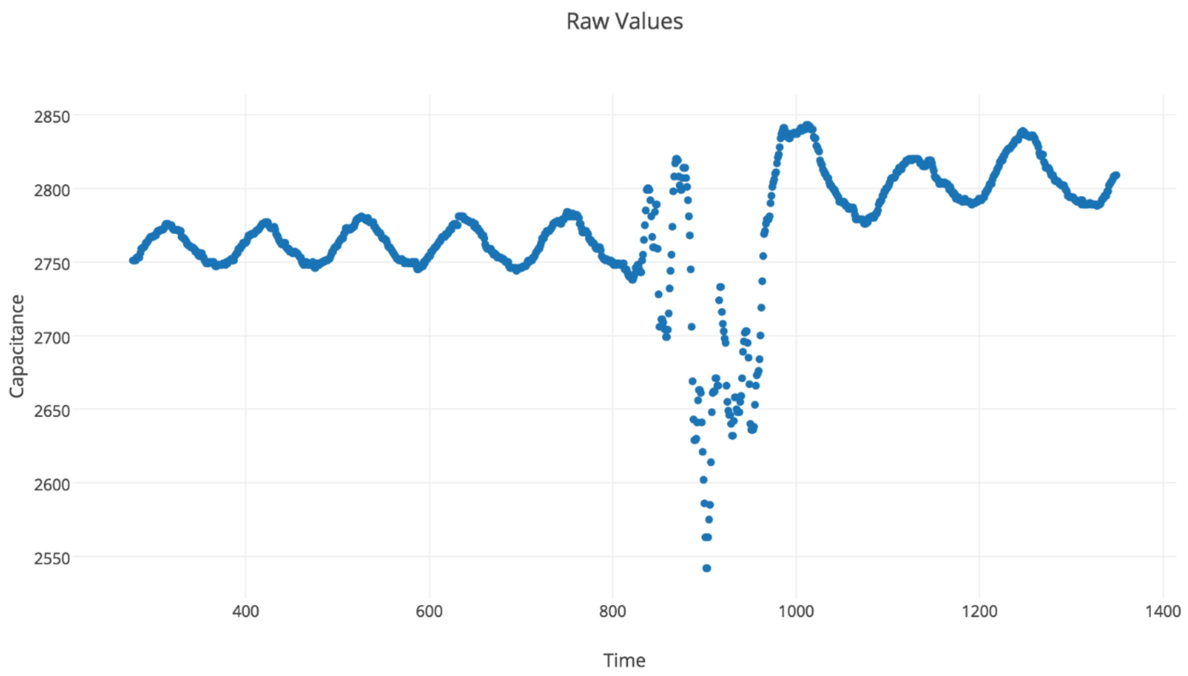
- **Identification:** Outliers can be detected using statistical methods such as Z-score, interquartile range (IQR), or visualizations like boxplots. More advanced approaches include machine learning–based anomaly detection.
- **Removal (Deletion):** Direct removal of outliers. Appropriate when outliers are due to measurement or data entry errors.
- **Transformation:** Apply mathematical transformations (e.g., logarithmic or square root) to reduce the impact of extreme values.
- **Imputation:** Replace outliers with summary statistics (mean, median, mode) or model-based estimates.
- **Capping:** Limit extreme values to predefined minimum and maximum thresholds.

Noise

In data analysis, **noise** refers to random variations, distortions, or errors that can occur during data collection, transmission, or storage.

These perturbations may arise from multiple sources, including human error, measurement instrument limitations, or natural variability in the observed phenomena.

Unlike outliers—which are extreme observations but not necessarily incorrect—**noise typically represents unwanted variability** that should be reduced or filtered to improve analytical accuracy.



Municipality	GDP (2018)
Joinville	30,785,682.00
Itajaí	25,413,431.00
Florianópolis	21,059,561.00
Blumenau	**9,587.83**
São José	10,607,482.00
Chapecó	9,602,336.00
Jaraguá do Sul	8,995,685.00
Criciúma	7,684,445.00
Brusque	6,375,501.00
Rio do Sul	**0.00**

Noise should be identified and corrected or filtered, as it can distort analysis and lead to misleading conclusions.

Unit 2 – Practical Assignment: Data Cleaning and Preparation

Objective

The purpose of this assignment is to operationalize core data preprocessing concepts through a structured workflow.

Students are expected to demonstrate the ability to justify dataset selection, formalize data structure through a data dictionary, and apply initial data cleaning techniques focusing on duplicates and missing values.

Assignment Tasks

1. Dataset Motivation and Contextualization

Provide a brief but analytically grounded justification for the selected dataset.

- Describe the domain and context of the dataset
- Explain its relevance (scientific, social, economic, or technical)
- Indicate potential analytical use cases (e.g., prediction, classification, descriptive analysis)
- Comment on dataset characteristics (size, structure, type: structured/semi-structured/unstructured)

Expected length: ~1 page

Assignment Tasks

2. Data Dictionary Construction or Adaptation

Develop or refine a structured data dictionary.

- List all variables/features in the dataset
- For each variable, specify:
  - Name
  - Description (semantic meaning)
  - Data type (e.g., numerical, categorical, datetime, text)
  - Units (if applicable)
  - Possible values or categories (for categorical variables)
- Identify potential data quality issues (e.g., ambiguous labels, inconsistent formats)

Format: Table (recommended: CSV, or spreadsheet)

Assignment Tasks

3. Initial Data Cleaning and Preparation

3.1 Duplicate Detection and Handling

- Identify duplicate records (exact and, if applicable, partial duplicates)
- Describe the detection strategy (e.g., full-row comparison, subset of columns)
- Report the number of duplicates found
- Apply and justify the chosen removal or consolidation method

3.2 Missing Data Treatment

- Quantify missing values (per variable and overall)
- Classify missingness patterns where possible (MCAR, MAR, MNAR – if inferable)
- Apply at least one treatment strategy:
    - ❑ Removal (listwise or column-wise)
    - ❑ Imputation (mean, median, mode, or simple heuristic)
- Justify the chosen approach in relation to the dataset and intended use

Deliverables

1. Short Report containing:
   - Dataset motivation and description
   - Explanation of the data dictionary
   - Description of cleaning procedures and decisions
   - Discussion of initial data quality issues

2. Data Dictionary File (CSV, Excel, or Markdown table)

3. Cleaned Dataset (CSV or equivalent), including:
   - Version after duplicate removal
   - Version after missing data treatment (can be the same file if sequential)

4. Code (optional but recommended):
   - Script or notebook (Python preferred) documenting the cleaning process
   - Should be reproducible and minimally documented